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Summary Physicist turned embedded systems engineer. Over ten years at Google building custom camera hardware and firmware for real-world imagery capture: FPGA soft-core firmware, precision time synchronization (IEEE 1588), high-speed sensor data paths, and the bench tooling that brings new hardware up. Ph.D. in physics with first-author papers in *Nature Communications* and *Optics Letters* and a patented instrument. Builds AI-agent-driven engineering workflows (MCP servers, autonomous coding agents) around real hardware.

Experience

Google LLC, Mountain View, California 2015–present

Software Engineer, Geo. Embedded and systems engineer on the team that designs and operates Google's custom camera platforms for capturing ground-level imagery for Google Maps.

- Own firmware for Xilinx FPGA-based camera systems: C/C++ on the MicroBlaze soft core, AXI-mapped peripherals (SPI, I^2C, MDIO, UART, GPIO), register maps, bitstream and ELF management, board bring-up, self-test, and validation over JTAG and UDP.
- Lead precision time synchronization and its validation across camera networks (IEEE 1588 PTP, hardware timestamping, transparent clocks). Designed an offline timeline-reconstruction estimator (Kalman filter with Rauch–Tung–Striebel smoothing and Allan-variance oscillator models) that maps free-running sensor timestamps onto a grandmaster UTC timeline; characterized oscillator drift and thermal behavior in deployed units.
- Built and maintain the team's camera development tool, a C++ CLI/TUI for hardware bring-up: register-tree and bus-topology browsing, self-test, usage telemetry, remote control over TCP/SSH, and an MCP server so AI coding agents can drive bench hardware directly. Packaged for Debian.
- Define wire formats and register layouts in Emboss (Google's open-source binary structure language) with generated C++ views; frame, metadata, and hardware-telemetry datagram protocols for 1G/10G Ethernet egress with DDR buffering, validated with capture-and-decode tooling.
- Replaced a vendor-specific Ethernet PHY driver with a spec-compliant MDIO Clause 45 driver with deterministic link bring-up, isolating device-specific code paths ahead of an I/O board respin.
- Introduced agent-driven development to the team: scheduled autonomous coding agents working a file-based issue queue behind test and lint gates, design-document review conventions, and local retrieval tooling over datasheets and register documentation.

Max Planck Institute for the Science of Light, Erlangen, Germany 2011–2015

Doctoral researcher, Laboratory of Nanophotonics and Biosensing (Dr. Frank Vollmer). Theoretical and experimental research on ultrasensitive label-free biodetection.

- Invented the centrifugal force quartz crystal microbalance, a force-based biosensor that probes a sample's mechanical properties non-destructively in real time (*Nature Communications* 2014; PCT patent application).
- Discovered and modeled a new interference mechanism in conically scattered surface plasmon light, then used plasmon speckle to resolve single-nanoparticle events in an unmodified SPR setup (*Optics Letters* 2013; *Nanotechnology* 2022).
- Designed and ran experiments with oligonucleotides, lambda DNA, self-assembled monolayers, and functionalized nanoparticles; built the optics, electronics, and data acquisition for each instrument. Supervised a master's thesis (Jiapeng Huang, 2014).

Friedrich-Alexander-Universität Erlangen-Nürnberg and **University of Oregon** 2009–2011

Master's researcher (Prof. Norbert Lindlein); guest researcher (Prof. Stephen Gregory). Modeled ultrashort pulses in the focal region of high-NA optics, with a parallel cluster implementation of the discrete Fourier transform; simulated surface plasmon polariton multiple scattering from apertureless near-field probes.

Light Beam Industries, Eugene, Oregon 2007–2009

Senior technician, R&D, optics and electronics. Designed digital, analog, and mixed-signal circuitry for power, control, and thermal management of LED optical products (fifteen documented PCBs, five in production, zero functional errors) and built a one-meter integrating sphere for photometry.

University of Oregon, Eugene, Oregon 2005–2007
Undergraduate research assistant (Prof. Daniel Steck). Built atom optics instrumentation: a one-meter scanning Michelson interferometer (1 in 10^6 resolution), low-noise high-speed photodiode detectors, a high-vacuum single-photon avalanche photodiode housing, and a tunable extended-cavity diode laser.

Education

Dr. rer. nat. (Ph.D.), Physics 2018
FAU Erlangen-Nürnberg and Max Planck Institute for the Science of Light, Erlangen, Germany
Dissertation: *Novel Methods in Optical and Mechanical Biosensors*

Master of Science, Theoretical Physics: Optics 2011
FAU Erlangen-Nürnberg, Erlangen, Germany. Thesis: *Ultrashort Pulses in Focal Regions*

Master of Science, Applied Physics: Optics 2009
University of Oregon, Eugene, Oregon

Bachelor of Science, Physics 2007
University of Oregon, Eugene, Oregon

Skills

- **Languages:** C, C++ (modern), Python, Bash (expert); TypeScript/JavaScript, MATLAB/Octave, Tcl, Starlark/Bazel, Perl, SQL, LaTeX (experienced); 6502 and MicroBlaze assembly (working).
- **Embedded and hardware:** Xilinx/AMD FPGA platforms (MicroBlaze, AXI, Vivado, bitstreams), JTAG debug, SPI/I²C/UART/MDIO, Ethernet PHYs and 1G/10G data paths, DDR buffering, IEEE 1588 PTP and hardware timestamping, image sensors and camera pipelines, Emboss, Linux kernel modules and DKMS, Debian packaging.
- **Modeling and signal processing:** Kalman filtering and smoothing, Allan variance and oscillator noise, Fourier optics, FDTD simulation (meep), finite-element analysis, PID control, MPI/CUDA parallel computing.
- **AI-assisted engineering:** Claude Code and MCP server design, autonomous agent workflows with test gates, local LLM inference, hybrid BM25/embedding retrieval (SQLite FTS5, ONNX), PyTorch.
- **Electronics and fabrication:** schematic and PCB design, SMT assembly and reflow, mixed-signal debug; manual and CNC machining (Haas, Bridgeport), CAD/CAM (Solid Edge, Mastercam), laser cutting, TIG/MIG/stick welding, spin coating, sputtering.

Publications

- Hongki Lee, Joel Berk, Aaron Webster, Donghyun Kim, and Matthew R. Foreman. "Label-free detection of single nanoparticles with disordered nanoisland surface plasmon sensor". In: *Nanotechnology* 33.16 (2022), p. 165502
- Aaron Webster, Frank Vollmer, and Yuki Sato. "Probing biomechanical properties with a centrifugal force quartz crystal microbalance". In: *Nature Communications* 5 (2014), p. 5284
- Aaron Webster and Frank Vollmer. "Interference of conically scattered light in surface plasmon resonance". In: *Optics Letters* 38.3 (2013), pp. 244–246
- Aaron Webster. "Novel Methods in Optical and Mechanical Biosensors". PhD thesis. Erlangen, Germany: Friedrich-Alexander-Universität Erlangen-Nürnberg, 2018

Patents

- Aaron Webster and Frank Vollmer. "Centrifugal force quartz crystal microbalance". Pat. req. WO 2015/042025 A1. PCT/US2014/055806, filed 16 September 2014. Mar. 26, 2015

Projects

- Open-source Emboss packet definitions for USB Power Delivery, UniFi, and cemuhook protocols.
- Patched Linux `hid-playstation` kernel driver exposing DualSense Edge paddles and function buttons, distributed via DKMS.
- Real-time PCM to Dolby Digital (AC-3) IEC 61937 encoder daemon in C; firmware analysis of a Dolby DP564 decoder (MPC860, dual DSPs) for a network audio tap.
- HALDO Labs: independent game studio publishing browser games in TypeScript and a 6502 assembly port for the Atari 800.

Service

- Director, Board of the Canyon Education Foundation 2026–present
- Volunteer technology lead, Canyon Elementary School District: network, Google Workspace administration, website and DNS migration, and ChromeOS fleet automation with the Google Admin SDK and Chrome Policy API 2021–present

Languages

English (native), German (CEF B1, conversational), Spanish (CEF A2).